

PART 1 GENERAL

1.1 SUMMARY

- A. Provide Airport Traffic Control Tower (ATCT) Cab Glass units of the highest quality and in a timely and expeditious manner for the replacement of defective glass units at various Federal Aviation Administration (FAA) and additional governmental and/or municipal aircraft control towers. Provide F.O.B. delivery of product(s) to the FAA Logistics (FAALC) at the Mike Monroney Aeronautical Center (MMAC) in Oklahoma City, Oklahoma, or directly to the ATCT site indicated in the delivery order. ATCT locations include the continental US, Alaska, Hawaii, Puerto Rico, and Guam.

1.2 REFERENCES AND STANDARDS

- A. American National Standards Institute (ANSI)
 - 1. ANSI Z 97.1 Safety Glazing Materials Used in Buildings - Safety Performance Specifications and Methods of Test
- B. American Society for Testing and Materials (ASTM)
 - 1. ASTM C 162 Standard Terminology of Glass and Glass Products
 - 2. ASTM C 1036 Standard Specification for Flat Glass
 - 3. ASTM C 1087 Standard Test Method for Determining Compatibility of Liquid-Applied Sealants with Accessories Used in Structural Glazing Systems
 - 4. ASTM C 1172 Standard Specification for Laminated Architectural Flat
 - 5. ASTM C 1376 Standard Specification for Pyrolytic and Vacuum Deposition Coatings on Flat Glass
 - 6. ASTM E 773 Standard Test Method for Accelerated Weathering of Sealed Insulating Glass Units
 - 7. ASTM E 774 Standard Specification for Sealed Insulating Glass Units
 - 8. ASTM E 1300 Standard Practice for Determining Load Resistance of Glass in Buildings
 - 9. ASTM E 1886 Standard Test Method for Performance of Exterior Windows, Curtain Walls, Doors, and Impact Protective Systems Impacted by Missile(s) and Exposed to Cyclic Pressure Differentials
 - 10. ASTM E 1996 Performance of Exterior Windows, Curtain Walls, Doors, and Impact Protective Systems Impacted by Windborne Debris in Hurricanes
 - 11. ASTM E 2188 Standard Test Method for Insulating Glass Unit Performance
 - 12. ASTM E 2189 Standard Test Method for Testing Resistance to Fogging in Insulating Glass Units
 - 13. ASTM E 2190 Standard Specification for Insulating Glass Unit Performance and Evaluation
 - 14. ASTM E 2461 Standard Practice for Determining the Thickness of Glass in Airport Traffic Control Tower Cabs

15. ASTM E 2649 Standard Test Method for Determining Argon Concentration in Sealed Insulating Glass Units Using Spark Emission Spectroscopy
 16. ASTM F 1642 / F 1642M Standard Test Method for Glazing and Glazing Systems Subject to Airblast Loadings
 17. ASTM F 2248 Standard Practice for Specifying an Equivalent 3-Second Duration Design Loading for Blast Resistant Glazing Fabricated with Laminated Glass
 18. ASTM F 2912 Standard Specification for Glazing and Glazing Systems Subject to Airblast Loadings
- C. Insulating Glass Manufacturers Alliance (IGMA)
1. IGMA TB-3001 Guidelines for Sloped Glazing
 2. IGMA TB-1601-95 Guidelines for Use of Capillary / Breather Tubes
 3. IGMA TM-3000 North American Glazing Guidelines for Sealed Insulating Glass Units for Commercial & Residential Use
 4. IGMA TM-4000-02 – Insulating Glass Manufacturing Quality Procedures
 5. IGMA TM-4100-03 – Preventing Insulating Glass Failures Manual
- D. Sealed Insulating Glass Manufacturers Association (SIGMA)
1. A1202 Commercial Insulating Glass Dimensional Tolerances
- E. Glass Association of North America (GANA)
1. Laminated Glazing Reference Manual
 2. Glazing Manual
 3. Sealant Manual
- F. Consumer Product Safety Commission (CPSC)
1. CPSC 16 CFR 1201 – Safety Standard for Architectural Glazing Materials

1.3 DEFINITIONS

- A. Manufacturer: Where used in this specification to refer to a firm that produces primary glass or fabricated glass as defined in the referenced standards.
- B. Fabricator: Where used in this specification to refer to a firm that fabricates glass units as defined in the referenced standards.
- C. Deterioration of Glass: Defects developed from normal use and weather conditions that are attributed to the manufacturing or fabricating process and not to causes other than glass breakage and practices for maintaining and cleaning glass contrary to the fabricator's directions. Defects including glass found to be out of compliance with ASTM C1036.
- D. Deterioration of Coated Glass: Defects developed from normal use that are attributed to the manufacturing or fabricating process and not to causes other than glass breakage and practices for maintaining and cleaning coated glass contrary to the manufacturer's directions. Defects include peeling, cracking, discoloration, and other indications of deterioration in the coating.

- E. Deterioration of Insulating Glass: Failure includes, but is not limited to, failure of the hermetic seal under normal and weather conditions due to causes other than glass breakage and improper practices for maintaining and cleaning insulating glass. Evidence of failure is the obstruction of vision by dust, moisture, film, or minerals on the surfaces of insulating glass facing the interspace or optical distortions (ghosting / double-images) not due to improper practices for maintaining and cleaning glass not in compliance with the manufacturer's or fabricator's directions.
- F. Deterioration of Laminated Glass: Defects developed from normal use and weather conditions that are attributed to the manufacturing or fabricating process and not to glass breakage, and practices for maintaining and cleaning laminated glass contrary to the manufacturer's or fabricator's directions. Defects include edge separation, delamination materially obstructing vision through glass, and blemishes exceeding those allowed by the referenced laminated glass standard. Laminated glass shall be considered defective when dissolved gases come out of solution (forming bubbles, haze, or delaminated areas between the glass/interlayer interface) by any cause, including elevated temperatures due to weather conditions and sunlight exposure.
- G. Glass Thickness: Designated in inches or millimeters according to ASTM C 1036 and E 2461.
- H. Interspace or Airspace: Air or inert gas-filled space between lites of an insulating glass unit.

1.4 SUBMITTALS FOR EVERY DELIVERY ORDER

- A. Product Data: Submit manufacturer's descriptive product data for each glass product and material used for glass unit fabrication, including coatings (if specified in the order), spacer, desiccant, primary sealant, and secondary sealants for insulating glass units, and the interlayers for laminated units. Submit manufacturer's handling and storage recommendations, installation instructions, and cleaning instructions from both the manufacturer of the glass lites and the manufacturer of the insulating glass and laminated glass units.
- B. Samples: Submit two 12" x 12" (305 mm x 305 mm) samples of the glass product ordered. Ship one sample to the FAALC and one to the ATCT, as indicated in the delivery order.
- C. Compatibility Certificates and Test Reports: Submit compatibility and adhesion certificates and test reports from the manufacturer of insulating glass edge sealant indicating that glass edge sealants were tested for compatibility with other glazing materials, including sealants, glazing tape, gaskets, setting blocks, and edge blocks. Include recommendations for glazing tape and caulking materials that are compatible with sealants and coatings used in assembled units.
- D. Product Certificates: Submit certificates that glass products comply with Insulated Glass Certification Council (IGCC), IGMA, Keystone, or other comparable IG certification programs and meet the performance requirements of ASTM E 2190 per test methods E 2188 and E 2189. If ordered unit(s) are for an application where the manufacturer recommends capillary tubes be left permanently open, the test report must indicate the product has a capillary tube left open during testing.

- E. User Information: Submit glass product handling and storage recommendations, installation instructions, and cleaning and maintenance instructions.

1.5 QUALITY ASSURANCE

- A. Fabricator must be approved and certified by the primary glass manufacturer for low-E coated glass and insulating glass units and must have successfully performed fabrication of Cab Glass units that meet FAA cab glass specifications under 5 contracts or orders for a federal agency within the past five years.
- B. The government reserves the right to visit on demand the fabricator's facility to examine glass products, production areas, operations, and their Quality Control (QC) Program, inspection process, and test reports.
- C. Fabricator must maintain IGCC Certification and provide updated documentation whenever changes are made, and test reports are received.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Deliver glass products to the destination indicated in the delivery order.
- B. Handle, store, and protect glass products in accordance with the manufacturer's written instructions.
- C. Prior to shipment, test all newly fabricated cab glass units for visual distortion. Perform test under all levels of light, with unit tilted in its' final 15° configuration, while viewing many types and shapes of objects, including moving objects and red and white lights. Visual ghosting or double imaging will not be acceptable.
- D. Prepare glass products for shipment in accordance with the manufacturer's written instructions using appropriate wood and glass blocking materials to prevent damage while in shipment from condensation, moisture ingress, temperature changes, direct exposure to the sun, or other causes. Allow sealants to cure fully prior to boxing, crating, and shipping.
- E. Box, crate, and transport all glass products in the appropriate vertical position exclusively.
- F. Provide temporary labels on each glass unit with the following information: date of fabrication, unit identification number, dimensions, weight, glass type and thickness, and spacer information if an insulating glass unit (IGU).
- G. Only when necessary for pressure equalization, and in accordance with manufacturers' written instructions, provide capillary tubes in IGUs exposed to substantial altitude changes between the location of fabrication and the destination site, or during transit to the site, to avoid hermetic seal rupture, or any extreme bowing inward or outward of the lites.

1.7 WARRANTY

- A. Insulating Glass Units
 - 1. Provide warranty signed by fabricator agreeing to replace insulating glass units that deteriorate as defined in Section 1.3 of this specification, FOB point

of manufacture, freight allowed project site, within 10 years after date of Final Acceptance, or within 5 years for units over 65 square feet or over 96 by 96 inches. Warranty covers only deterioration due to normal conditions of use as defined and not due to handling and installing.

B. Laminated Glass Units

1. Provide warranty signed by fabricator agreeing to replace laminated glass units that deteriorate as defined in Section 1.3 of this specification, FOB point of manufacture, freight allowed project site, within 10 years after date of Final Acceptance. Warranty covers only deterioration due to normal conditions of use as defined and not due to handling and installing.

C. Coated Glass Products

1. Provide warranty signed by fabricator agreeing to replace coated glass units that deteriorate as defined in Section 1.3 of this specification, FOB point of manufacture, freight allowed project site, within 10 years after date of Final Acceptance. Warranty covers only deterioration due to normal conditions of use as defined and not due to handling and installing.

PART 2 PRODUCTS

2.1 GLASS MANUFACTURERS

- A. Subject to strict compliance with this specification, provide glass from one of the following:
1. Vitro / PPG
 2. Nippon / Pilkington
 3. Saint-Gobain
 4. Guardian
 5. AGC Glass

2.2 GLASS PRODUCTS

- A. All glass must meet ASTM C 1036-11, Type I (Transparent Glass, Flat), Class I (Clear), and Q2 (Quality Q2), as well as this revision in Table 3: "Note A shall not apply; instead, the largest point blemish in glass ¼" to ½" (6 mm to 12 mm) thick shall be smaller than 0.060" (1.5 mm)."
- B. Coated glass must meet ASTM C1376, coated by vacuum deposition (sputter-coating) process, and subject to compliance with this specification.
- C. All glass fabrication shall be vertical-assembly and automated. Etching or permanent labels on glass are not acceptable, neither are scratches on any glass surface.

2.3 LAMINATED GLASS

- A. Laminated Glass Units: Provide units in accordance with ASTM C1172 and this specification.
- B. Interlayer: Clear and ultra-clear polyvinyl butyral (PVB) sheets in compliance with

ASTM C1172 and GANA Laminate Manual. Interlayer shall be clear or ultra-clear as indicated in the delivery order, and thickness of not less than 0.060" (1.5 mm). Use materials that have a proven record of no tendency to bubble, discolor, or lose physical and mechanical properties after fabrication and installation. Maximum allowable edge-overlap of the plies to be 0.12" (3 mm) for butt joints and 0.2" (5 mm) for all other edges. Comply with the manufacturer's written instructions for edge deletion. When ordered, provide clear rigid ionomeric polymer interlayer with structural properties to achieve wind load resistance as required. Unless otherwise approved by the FAA, manufacturers shall be:

1. Kuraray / Trosifol / SentryGlas
 2. Eastman / Safeflex / Vanceva
- C. Edgework: Post-laminated CNC machine ground flat and polish glass edges and corners, with slight chamfers on each side. For butt-glazed monolithic and laminated glass, CNC machine flat- grind vertical edges to produce mitered edges with slight chamfers at junctions of edges and faces.

2.4 INSULATING GLASS

- A. Insulating Glass Units: Provide units in accordance with this specification and qualified according to ASTM E2190.
- B. Hermetically Sealed System: Dual seal, with polyisobutylene primary and black silicone secondary sealants. Both seals must be continuous around spacer and maintain a constant depth to ensure a watertight and airtight seal and not impact the performance of the IGU. Any voids and skips in sealants will result in rejection of material. Undulations in the edge of the sealing system may be discernible but will be considered minor and not constitute rejection when the reduced depth of the seal is less than 10% reduction in depth. When indicated in the delivery order, increase sightline dimension up to 1.25" (32 mm) to provide additional IGU secondary sealant.
- C. Spacer: Spacers with plugged holes will result in rejection of material. Spacers shall meet the requirements below:
1. Material: Stainless steel, mill finish and black
 2. Corners: All corners shall be machine bent without cuts and corner keys; must be smooth and contain no out-of-plane bends on the surface which contacts the glass
 3. Joints: Bent corners with only one assembly joint permitted, located on the left or right side of the unit at mid-height of the unit
 4. Cuts & Splices: No burrs or jagged edges allowed
 5. Labels: Provide clearly visible label with date stamp
- D. Desiccant: Molecular sieve or silica gel, filled on all four sides. Use of desiccant prefill-in foam spacer must be submitted for approval and meet initial submittal documentation requirements with the justification for the substitution.
- E. Interspace: The hermetically sealed air space shall be dehydrated air, or when ordered, argon gas at a minimum concentration of 90%.
- F. Edgework: CNC machine ground flat and polish glass edges and corners, and with slight chamfers on each side.
- G. Low-E Coating: When ordered, coating material must be submitted for approval and

designed for the intended environment. Do not apply to glass thicker than $\frac{1}{4}$ " (6 mm). Apply to glass Surface 2 unless indicated otherwise in the delivery order. Provide edge deletion according to the manufacturer's recommendations to improve spacer adhesion and avoid coating deterioration.

- H. Capillary Tube: When recommended by the manufacturer for pressure equalization, provide a stainless steel capillary tube 12" (305 mm) long, 0.25 to 0.5 mm inside diameter, and install within the bottom half of a vertical edge in each of the units with the exposed end pointing downward and temporarily secured in this position during shipment in accordance with IGMA TB-1601-14.

LIST of ACRONYMS and ABBREVIATIONS:

ASCE – American Society of Civil Engineering ASTM – American Society for Testing and Materials
ATC – Airport Traffic Control
ATCT – Airport Traffic Control Tower Cab – An ATCT cabin
CO – Government Contracting Officer
COR – Government Contracting Officer’s Representative FAA – Federal Aviation Administration
FOB – Freight on Board IG – Insulating Glass
IGCC – Insulating Glass Certification Council IGU – Insulating Glass Unit
Low-E – low emissivity glass mm -- millimeter
PVB – Polyvinyl Butyral QA – Quality Assurance
QA/QC – Quality Assurance/Quality Control SOW – Statement of Work